

Global CO₂ Emissions: Trends, Regional Shifts, and Per Capita Inequalities

A data-driven overview of global fossil fuel and industrial CO₂ emissions (1750–2024)

Scope

- 1750–2024 emissions trajectory
- Regional responsibility shifts
- Per capita inequality and 2024 change

Global CO₂

Policy Briefing

Atmospheric CO₂ Reaches 429 ppm — 50% Above Pre-Industrial Levels

429 ppm

February 2026
Mauna Loa (NOAA)
~280 ppm pre-industrial baseline

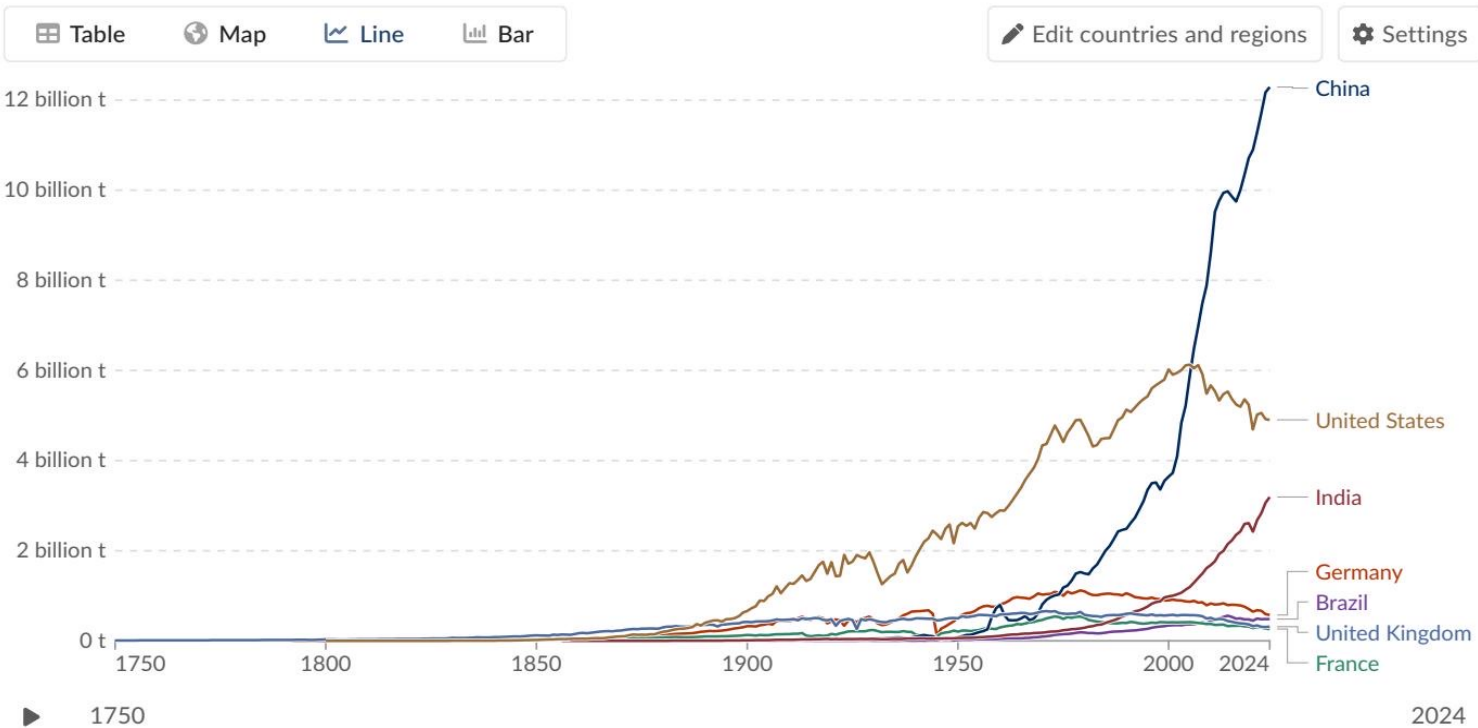
CO₂ concentration is now at least 1.5× greater than the natural rise seen after the last ice age — but compressed into ~200 years.



Global Fossil Fuel CO₂ Emissions Grew Six-Fold: 6 Billion to 35+ Billion Ton

Annual CO₂ emissions

Carbon dioxide (CO₂) emissions from fossil fuels and industry. Land-use change emissions are not included.



Data source: Global Carbon Budget (2025) – [Learn more about this data](#)

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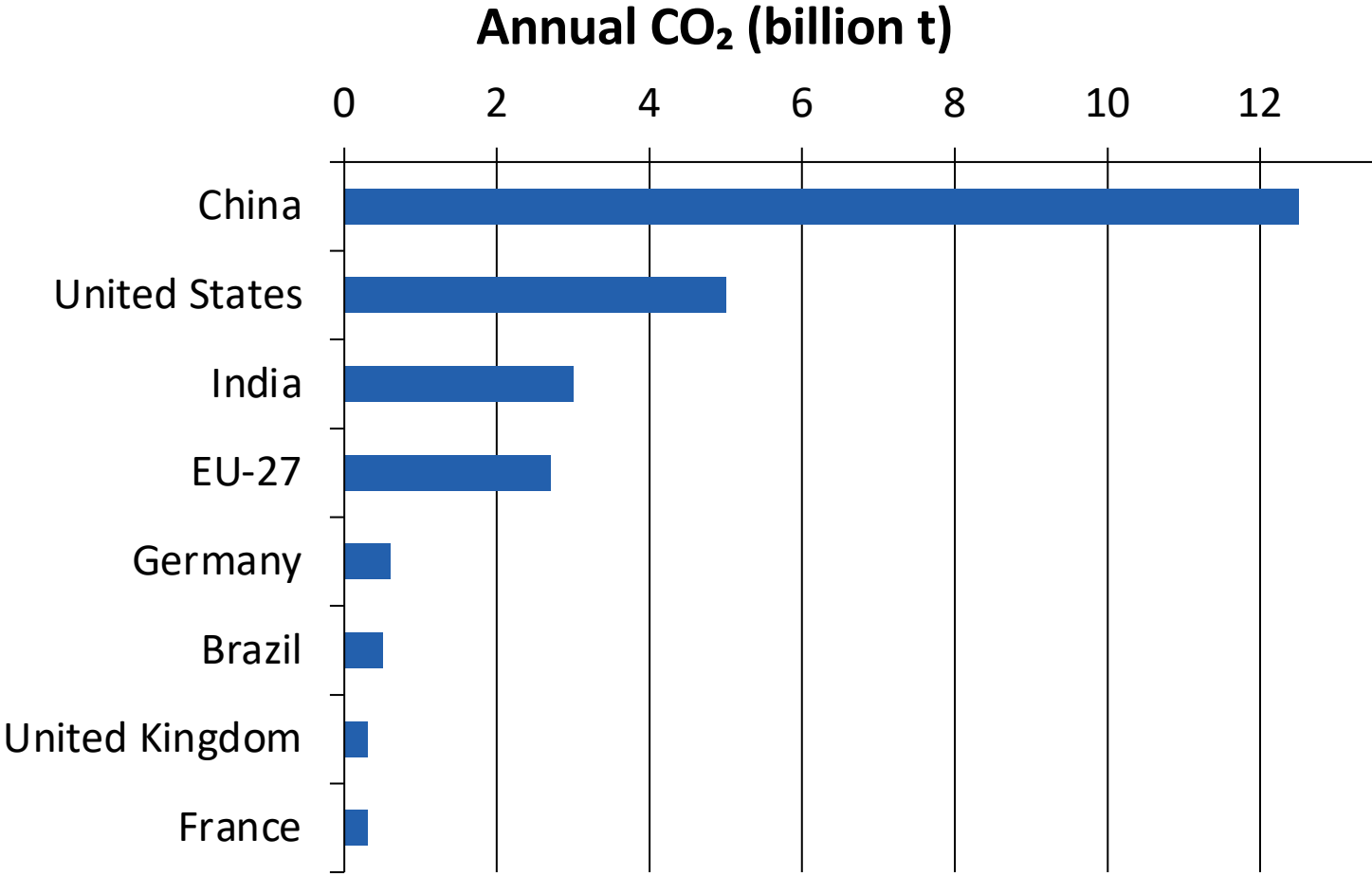
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Key inflection points

- 1950: ~6 billion tonnes
- 1990: >20 billion tonnes
- 2024: >35 billion tonnes
- Growth slowed, but no peak yet

Reference visual: figure_3.jpg (country trajectories shown in source brief)

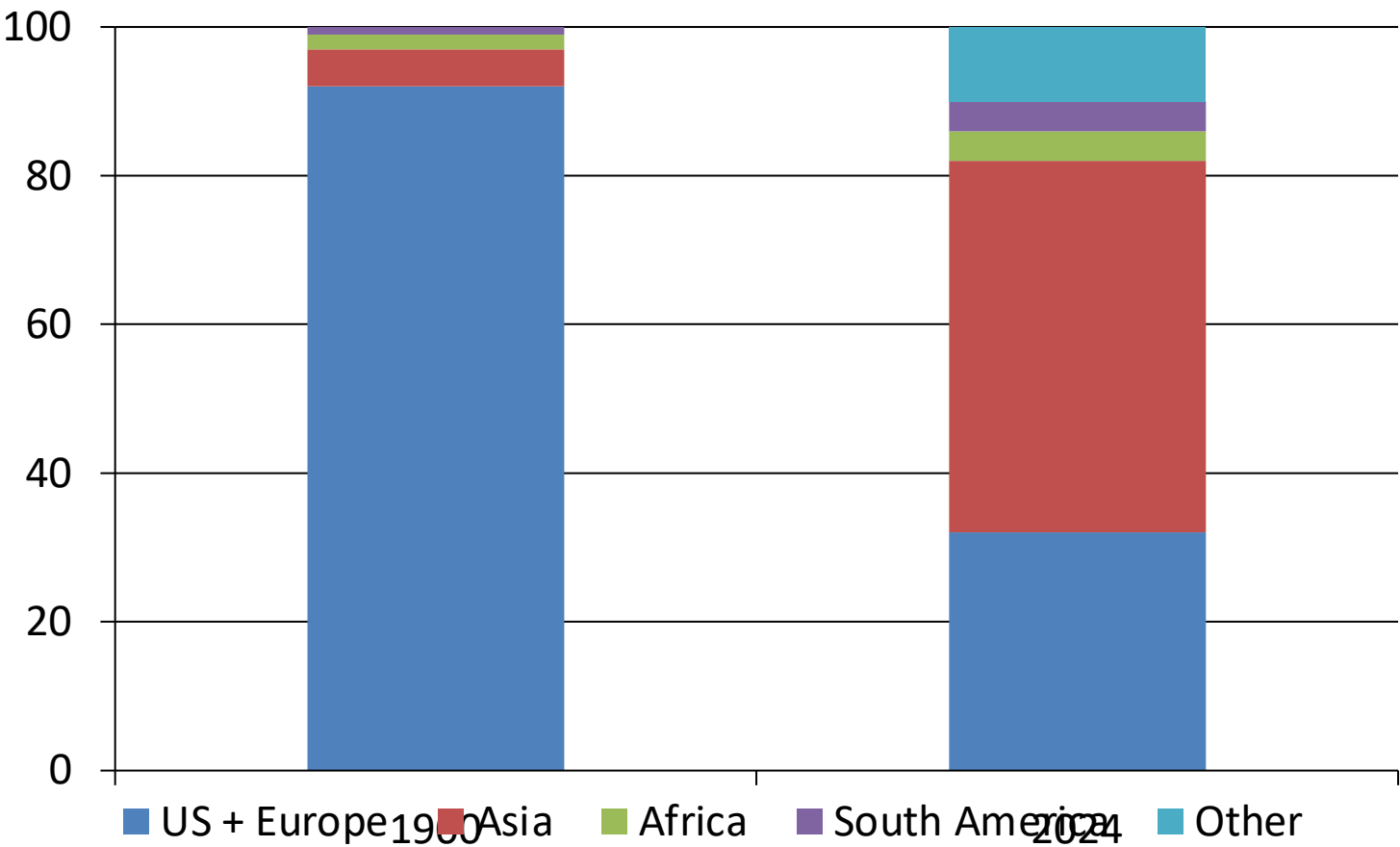
China Now Emits Over 12 Billion Tonnes — More Than Double the United



Analytical points

- China: ~27% of global total
- US emissions down from ~6 Bt peak
- India rising to ~3 Bt
- European totals declining

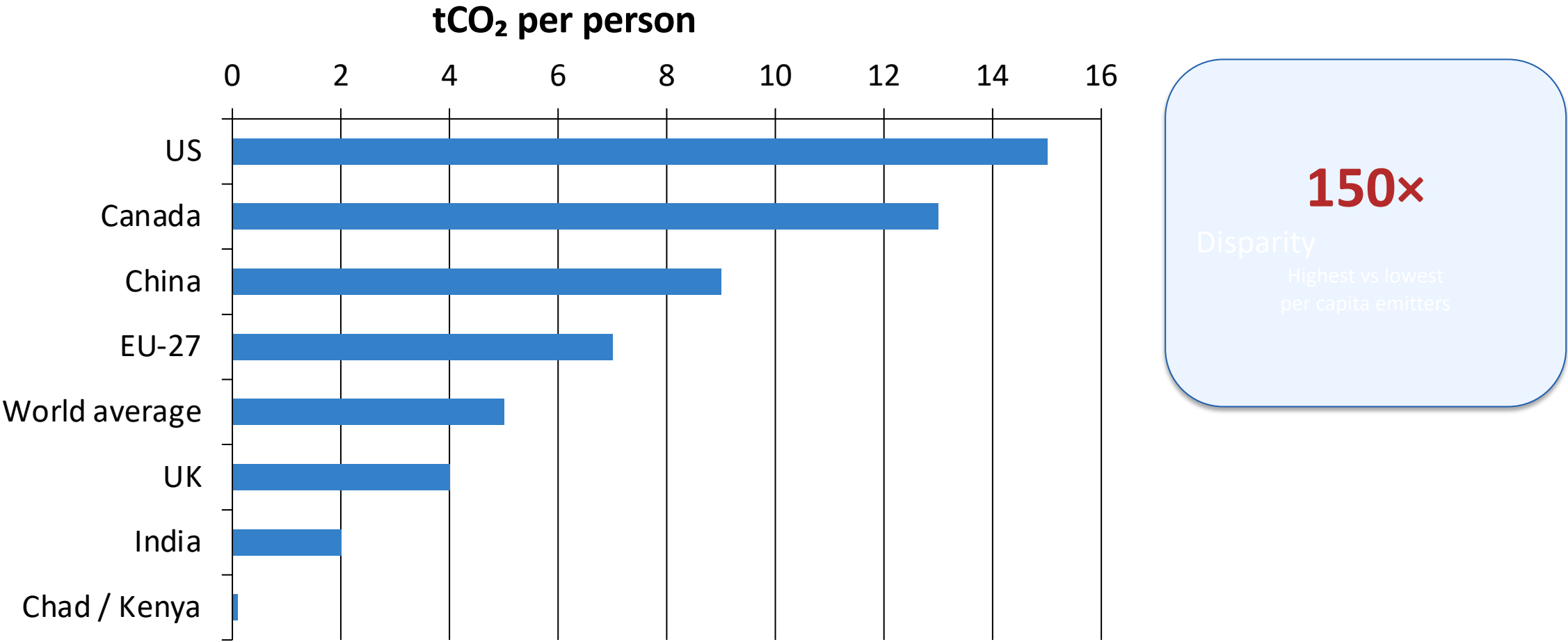
Asia Now Accounts for ~50% of Global Emissions; US + Europe Below One



Shift summary

- 1900: Europe+US >90%
- 2024: Asia ~50%
- US+Europe now < one-third
- Africa and South America each ~3-4%

Per Capita Gap: 15 t in the US vs. 0.1 t in Chad — a 150× Disparity



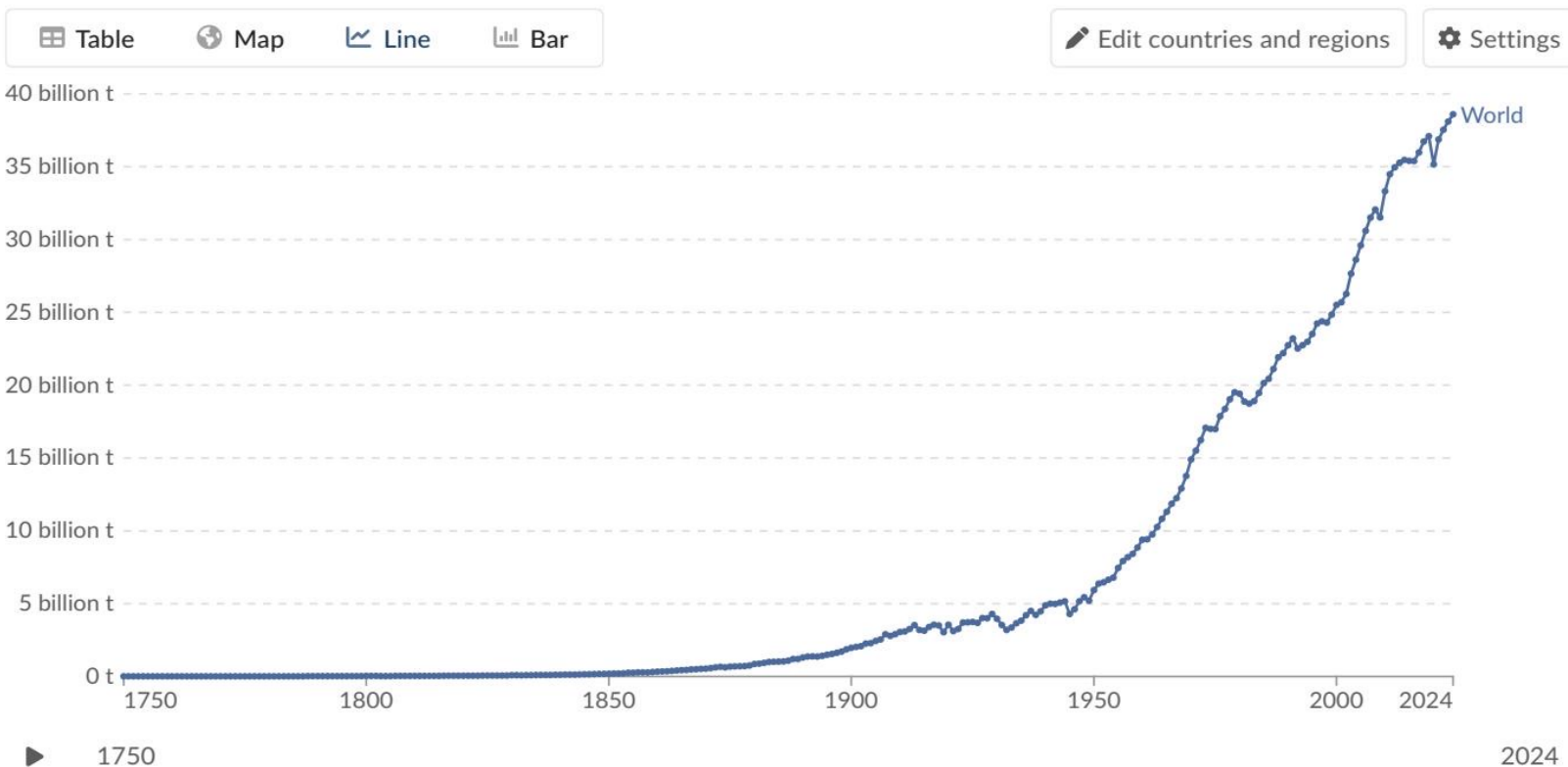
Reference trend context: figure_1.jpg

2024 Snapshot: US and Europe Cut Emissions While Russia and SE Asia Inc

Annual CO₂ emissions

Carbon dioxide (CO₂) emissions from fossil fuels and industry. Land-use change emissions are not included.

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Data source: Global Carbon Budget (2025) – [Learn more about this data](#)
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2024 pattern

- US + much of Europe: –1% to –5%
- Russia and SE Asia: +2% to +10%
- Parts of Africa: increases
- Mixed outcomes in South America

Energy Mix Explains Why France Emits Far Less Per Capita Than Germany

Countries with similar prosperity can have very different CO₂ outcomes depending on electricity and heat sources.

Country group	Dominant energy mix	Per capita CO ₂ (t/person)	Implication
France / UK	Higher nuclear + renewables	~4	Lower per capita emissions
Germany / Netherlands	Higher fossil fuel share	Higher than France/UK	Higher per capita emissions
Policy signal	Power-sector choices matter	—	Energy mix shifts national footprint
Negotiation relevance	Same income, different intensity	—	Supports differentiated pathways

Key Takeaways: Emissions Are Still Rising, But the Map of Responsibility Is

- 1) Global emissions exceed 35 billion tonnes and have not yet peaked.
- 2) China emits over 25% of the global total (>12 Bt), more than double the US.
- 3) Per capita inequality remains extreme: ~150× between highest and lowest emitters.
- 4) US and Europe are declining, but historical cumulative responsibility remains large.
- 5) Policy and energy-mix choices—not prosperity alone—shape per capita outcomes.